

MADELEINE CORTES SANTOS

Scientific Assistant/PhD Student

PERSONAL PROFILE

Born: April 11, 1999 (26 y/o)
Place of birth: Tarlac, Philippines
Citizenship: Filipino

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HIGHLIGHTS

- Currently employed as a scientific assistant/PhD student at ETH Zurich, Switzerland
- Worked as a student assistant while obtaining a master's degree at ETH Zurich for various projects applying organic and isotope geochemistry and data management
- Previously employed as an Instructor of geology undergraduate classes in University of the Philippines
- Worked as a Research Assistant for an energy research project in the Philippines
- Worked as an international research intern in Academia Sinica (Taiwan International Graduate Program)
- Knowledgeable in Python, MATLAB, and R with training in machine learning applications for earth sciences

ACHIEVEMENTS

- Awarded the ETH Medal for Outstanding Master's Thesis and Prize of the Zurich Geological Society for an excellent Master's thesis ("Preis der geologischen Gesellschaft Zürich für exzellente Masterarbeit")
- Awarded the Excellence Scholarship and Opportunity Program from ETH Foundation from 2021-2023
- Graduated in Bachelor's (Geology) at the University of the Philippines *with honors* in 2019
- Awarded the Department of Science and Technology (DOST) Merit Scholarship from 2015 to 2019

ACADEMIC HISTORY

Swiss Federal Institute of Technology (ETH Zurich), Canton of Zurich, Switzerland

Doctor of Sciences in Earth Sciences (Nov 2024-present), in progress

Extracurricular activities and organizations: Association of Scientific Staff at ETH (AVETH) commissioner for the Excellence Scholarship and Opportunity Programme

Swiss Federal Institute of Technology (ETH Zurich), Canton of Zurich, Switzerland

Master of Science in Earth Sciences major in Geology (Sept 2021-Aug 2023), with distinction

Thesis title: "Tracking Holocene sea ice and paleoenvironment evolution in the Beaufort Sea, Arctic Ocean"

Studies fully-funded by the Excellence Scholarship and Opportunity Programme (CHF 48'000 for the duration of the master's program)

Relevant courses: Climate History and Paleoclimatology, Global Carbon Cycle-Reduced, Sedimentology

National Institute of Geological Sciences (NIGS), University of the Philippines - Diliman (UP Diliman)

Bachelor of Science in Geology (Aug 2015- June 2019), cum laude (with honors)

Studies funded by the Department of Science and Technology (DOST) Merit Scholarship

Extracurricular activities and organizations: University of the Philippines Department of Science and Technology Scholars' Association (executive committee 2018-2019), University of the Philippines Weightlifting Club, Dancesport Society, University of the Philippines Repertory Company

TRAINING & COMPETENCIES

- ★ Has experience in using various mapping and remote sensing software such as:
 - QGIS
 - ArcGIS
 - Global Mapper
 - Surfer
 - GMT
 - ENVI
 - LeapFrog
- ★ Has training in using Python, MATLAB, R
- ★ Has participated in fieldworks (e.g., geologic mapping, soil and water sampling, etc.)
- ★ Has experience in using geophysical software such as: 1) SonarWiz, and 2) Oasis Montaj
- ★ Knowledgeable in photo, video editing and image processing software such as Adobe Lightroom and imageJ
- ★ Trained in using equipment for geochemical analyses applying: 1) clumped isotope sample preparation and analysis, 2) total lipid extract extraction, purification and analysis in gas chromatography flame ionization detection and mass spectrometry, 3) X-Ray diffraction and fluorescence
- ★ Attended the Urbino Summer School in Paleoclimatology (16 days of lectures, symposia, and excursions on different topics in paleoclimatology including biogeochemical cycling, paleoceanography and deep-time climate modeling)

WORK EXPERIENCE

Project Assistant/PhD student

ETH Zurich, Switzerland | Nov 2024-Present

- Generating temperature and CO₂ concentration proxy records from marine sediments using various geochemical analyses (carbonate clumped isotope and alkenone analyses) and comparison to climate model output (CESM2)
- Developing an image classification program for recognizing marine algae carbonate plates (coccoliths) in microscope slide photos applying Residual Neural Network and Support Vector Machine
- Supervising master's thesis and assisting in teaching *Molecular Paleontology and Micropaleontology* and *Climate History and Paleoclimatology* courses (master's level courses)

Project Assistant

University of Basel, Switzerland | Oct 2023-Oct 2024

- Investigated methane concentration variability for the past 12,000 years in the Arctic using organic geochemistry

Hilfsassistent (Graduate student assistant)

Geological Institute, ETH Zurich | Jun 2022-Mar 2023, May 2023-Sept 2023

- Performed organic geochemical analyses (from lipid extraction to gas chromatography and mass spectrometry analysis) on sediments for Dr. Cindy de Jonge
- Performed silica extractions on marine sediments, coccolithophore size identification (using the image processing toolbox of MATLAB), and various laboratory work for Prof. Dr. Heather Stoll

Instructor III

National Institute of Geological Sciences, UP Diliman | Sept 2020 - August 2021

- Taught and prepared course materials for Principles of Geology, Earth History, and Elementary Petrology lecture and laboratory classes; also headed the junior faculty in preparing materials for the undergraduate Field Geology course amidst the pandemic

Project Research Assistant IV

National Institute of Geological Sciences, UP Diliman | Jan 2020 - Sept 2020

- Prepared technical reports, geologic/geophysical maps, figures and diagrams
- Processed and interpreted geophysical data
- Conducted petrographic and geochemical analyses

Museum Docent (volunteer)

National Institute of Geological Sciences, UP Diliman | Aug 2019 - Jan 2020

- Conducted museum lectures and curated materials for exhibits

Research Intern

Isotope Composition/Planetary Science Laboratory, Institute of Earth Science Academia Sinica, Taiwan | Jul-Aug 2019

- Conducted a research project involving sulfur isotopes, applying mass spectrometry and laser fluorination techniques

PUBLICATIONS and other scientific works

- Santos, M., Broeder, L., O'Regan, M., Hernandez-Almeida, I., Tesi, T., Bigler, L., Haghipour, N., Nelson, D., Fritz, M., Lattaud, J. (2026). Holocene sea ice and paleoenvironment conditions in the Beaufort Sea (Canadian Arctic) reconstructed with lipid biomarkers. *Climate of the Past* Volume 22. <https://doi.org/10.5194/cp-22-187-2026> (journal article)
- Maglalat, E., Armada, L., Santos, M., Sayen, K., Dimalanta, C., Hsu, S., Yumul, G. (2023). Bottom simulating reflectors in the Manila Trench forearc and its implications on the occurrence of gas hydrates in the region. *Marine and Petroleum Geology* Volume 158, Part A. <https://doi.org/10.1016/j.marpetgeo.2023.106538>. (journal article)
- Zhang, H., Huang, Y., Wijker, R., Cacho, I., Torner, J., Santos, M., Kost, O., Wei, B., and Stoll, H. (2023). Iberian Margin surface ocean cooling led freshening during Marine Isotope Stage 6 abrupt cooling events. *Nature Communications* 14, 5390 (2023). <https://doi.org/10.1038/s41467-023-41142-8> (journal article)
- Santos, M., Lattaud J., Hernandez-Almeida, I., Broeder, L. (2023). Tracking Holocene sea ice and marine paleoenvironment evolution in the Beaufort Sea, Arctic. *scientific poster for Goldschmidt 2023*)
- Santos, M., Liang, M.C. (2019). Development of multiple sulfur isotope analysis by means of fluorination and SF₆ dual-inlet gas-source mass spectrometry using IAEA Ag₂S standards, (scientific poster in Academia Sinica Taiwan International Graduate Program-International Internship Symposium)
- Santos, M., Magalona, L., Salinas, B.S. (2019) Petrography and geochemistry of Neogene Sandstones from the Bojeador Formation and Batac Formation of Central Ilocos Sur, Philippines: Implications for provenance and tectonic setting (paper and poster, presented in 16th Annual Meeting of Asia-Oceania Geosciences Society)